PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

KalwaPutra Edu

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By	Roll No.	Supervised By	Organization
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"KalwaPutra Edu"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr.Rajat Takkar** The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Aniket Kumar**

Date: 30-04-2026 Roll No2211981054

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

KalawatiPutra Edu is a unified ed-tech platform designed for college students to streamline technical preparation and career development. It solves the problem of fragmented resources by integrating DSA practice, structured courses, progress tracking, and career services into a single platform.

The system provides **pattern-wise and topic-wise DSA preparation**, advanced filtering (difficulty/company), and personalized features like daily goals, progress tracking, and streaks to ensure consistent learning.

It also includes **courses, roadmaps, mentorship, internship/placement listings**, and a **blog platform** for community engagement. A key highlight is the **AI-powered resume screening**, which generates ATS scores and interview questions. Additional features include certificate generation and a student discount store.

Built using a **scalable MERN stack**, the platform offers a complete ecosystem that enhances students' technical skills and career readiness in one place.

TABLE OF CONTENTS

S.No.	Title	Page No.
	Abstract	i
	Acknowledgements	ii
1.	Introduction	1
1.1	Problem Background	1
1.2	Objective of the Project	2
1.3	Proposed Solution / System Overview	2
1.4	Key Features or Functionality	3
1.5	Technology / Method Used	4
1.6	Expected Outcome & Benefits	4
1.7	Original Contribution Statement	5
2.	Identified Research Gap	5
3.	Software Requirement Specification (SRS)	6
4.	System Design and Architecture	6
4.1	System Architecture	7
4.2	Module Description	7
5.	Implementation Details	8
5.1	Technology Stack Description	8
6.	Results and Screenshots	9
7.	Conclusion and Future Scope	15
7.1	Conclusion	16
7.2	Future Scope	18
8.	References	20
9.	Appendices	22

1. INTRODUCTION

1.1 Problem Background

Students face difficulty due to **multiple disconnected platforms** for DSA, courses, placements, and resumes.
This leads to:

- Frequent context switching
- No unified progress tracking
- Lack of personalization
- Limited access to career tools

1.2 Objective of the Project

To build a **unified ed-tech platform** that:

- Combines learning, practice, and placement
- Provides smart tracking and goal setting
- Integrates career support (resume, mentorship, jobs)
- Offers additional services like certificates and student store

1.3 Proposed Solution / Overview

A **web-based integrated platform** where students can:

- Practice DSA with filters and progress tracking
- Follow structured courses and roadmaps
- Use AI-based resume analysis and interview prep
- Access blogs, certifications, and student store
- Manage everything in one place

1.4 Key Features

- DSA practice (topic, pattern, company-wise)
- Daily goals C progress tracking
- Courses + roadmaps
- Resume screening + AI interview questions
- Internship C placement listings
- Blog system + user profiles
- Certificate generation + student store
- Admin dashboard

1.5 Technology Used

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Cloud:** Cloudinary, Vercel, Hostinger
- **AI:** Gemini / OpenAI API
- **Tools:** GitHub, VS Code, Notion

1.6 Expected Outcome

- Better learning consistency
- Strong DSA + practical knowledge
- Improved placement readiness
- All-in-one platform → saves time C effort

1.7 Original Contribution

A **single integrated ecosystem** combining:

- Learning + Practice + Career + Commerce

- AI-powered resume analysis + interview generation
- Personalized tracking system

Feature Area	LeetCode/Gf G	Coursera/Ude my	LinkedIn	KalawatiPutra Edu (Proposed)
DSA Practice	Excellent	None	None	**Excellent**
Structured Courses	Basic/Limited	Excellent	None	**Good**
Placement/Internship Listings	None	None	Excellent	**Good**
Mentorship/Guidance	Forum-based	QCA Limited	Networking	**Integrated Module**
Resume Screening s Feedback	None	None	Basic Profile	**AI-Powered Analysis**
Personalized Goal Setting	Basic	Limited	None	**Comprehensive**
Student-Focused Commerce	None	None	None	**Integrated Store**
Unified Ecosystem	No	No	No	**Yes**

2.3 Identified Research Gap

The literature and market survey clearly identifies a significant gap: there is no single platform that holistically serves the complete lifecycle of a college student's technical and professional development. Students are forced to become "integrators," manually cobbling together an experience from disparate tools. This leads to inefficiency, lack of personalization, and a fragmented view of their own progress. KalawatiPutra Edu aims to fill this gap by building a truly unified ecosystem where learning, practice, career preparation, and student services coexist and interact intelligently.

3. SYSTEM MODELS (UML Diagrams)

User Flow

- User → Login/Signup → Dashboard
- From Dashboard, user can access:
    - **DSA Module:** Solve questions, track progress, maintain streak
    - **Courses:** Learn modules, earn certificates
    - **Resume Analyzer:** Upload resume, get ATS score C AI questions
    - **Blogs:** Create C read posts
    - **Store:** Purchase student products
    - **Mentorship:** Placement C guidance support

Admin Flow

- Admin → Admin Dashboard
- Admin can manage:
    - Users
    - Courses
    - Products
    - Blogs and Also handles analytics and full system control

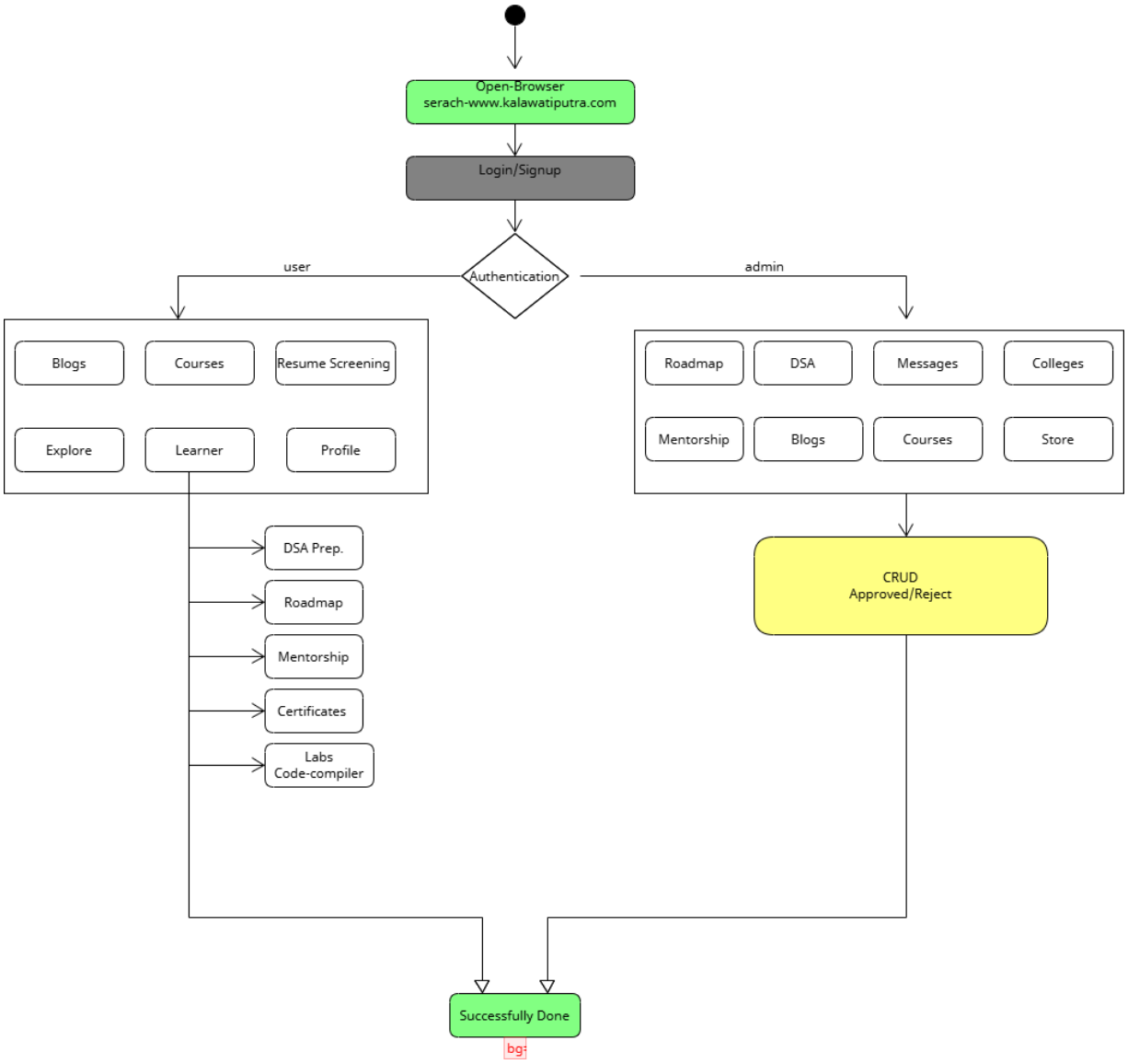
System Architecture Flow

- **Frontend → Backend (Node.js/Express)**
- **Backend → MongoDB Database**

External Integrations

- **Cloudinary:** File storage
- **Gemini/OpenAI API:** AI features
- **Nodemailer:** Email services

Workflow Diagram



4.SYSTEM DESIGN AND ARCHITECTURE

4.1System Architecture

KalawatiPutra Edu uses a **MERN-based client-server architecture** for scalability and efficiency.

- **Client Layer (React.js):**
  Handles UI, user interaction, and communicates with backend via APIs.
- **Server Layer (Node.js s Express.js):**
  Manages business logic, APIs, authentication (JWT), and data validation.

- **Database Layer (MongoDB Atlas):**
  Stores user data, courses, questions, blogs, and other content.
- **External Services:**
  - Cloudinary → Image storage
  - Gemini API → Resume analysis C question generation
  - Nodemailer → Email services
  - Google OAuth → Social login

4.2Module Description

- **User Module:** Registration, login, profile, role-based access
- **DSA Module:** Question filtering + user progress tracking
- **Learning Module:** Courses, roadmaps, and progress tracking
- **Career Module:** Resume upload, AI analysis, ATS scoring
- **Blog Module:** Create, update, delete blog posts
- **Commerce Module:** Store management + certificate generation
- **Admin Module:** Full control dashboard for managing platform data

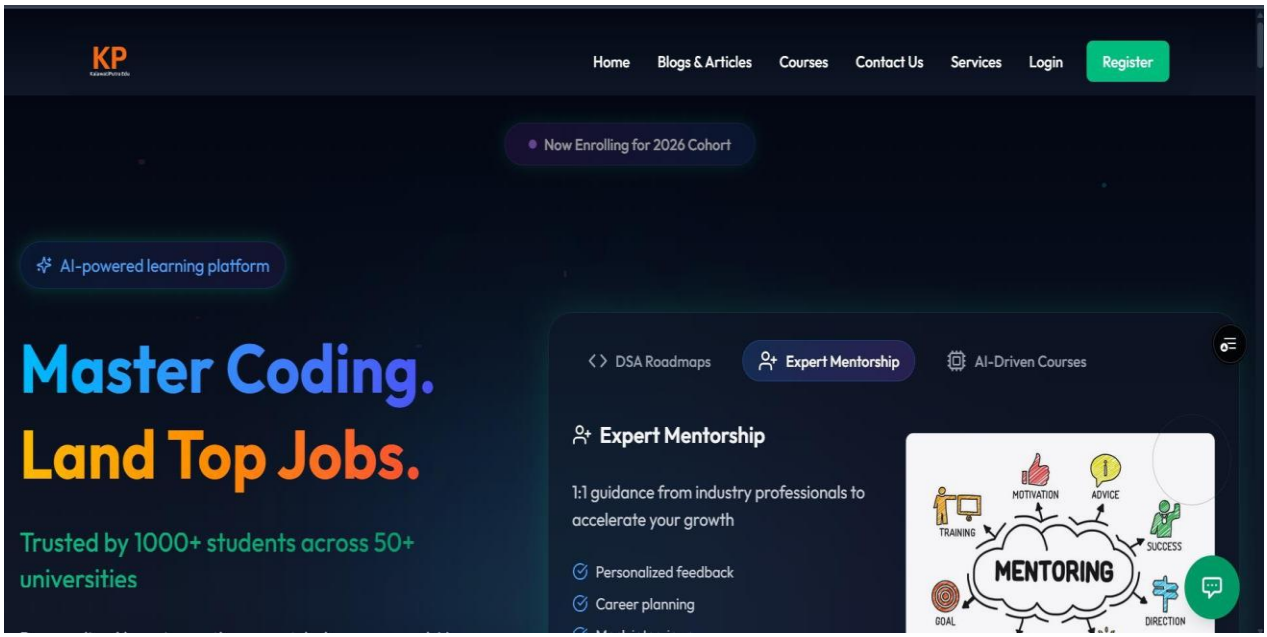
5.IMPLEMENTATION DETAILS

5.1Technology Stack Description

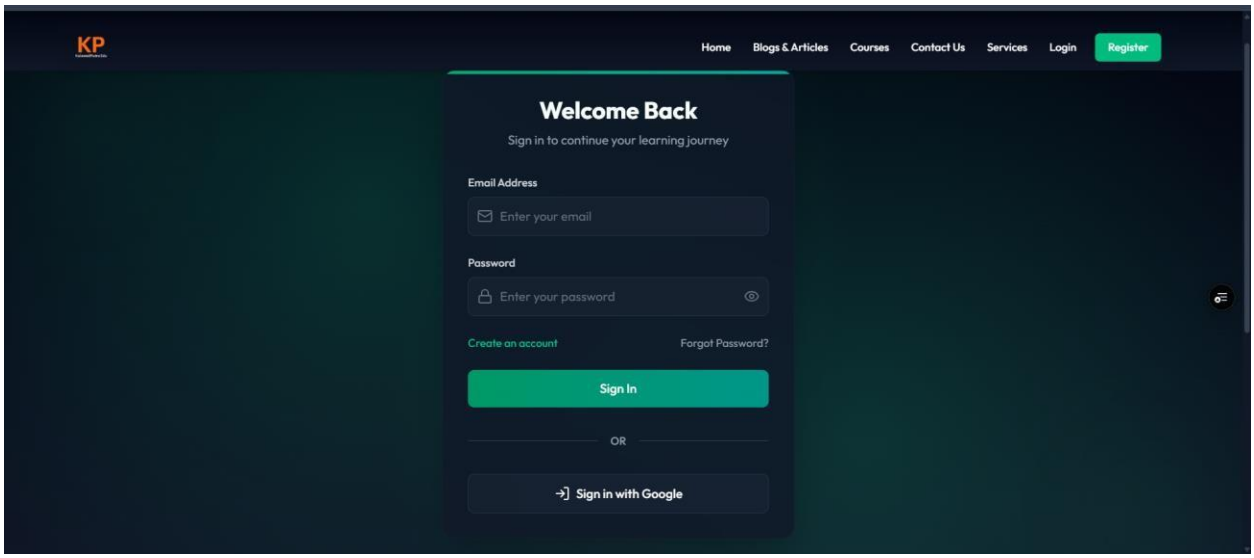
- **Frontend: React.js**
  Used for building a fast, dynamic, and reusable UI with component-based architecture.
- **Backend: Node.js s Express.js**
  Handles APIs efficiently using a non-blocking, scalable server architecture.
- **Database: MongoDB (with Mongoose)**
  Flexible NoSQL database for handling different data types with structured schema support.
- **Authentication: JWT s Bcrypt**
  JWT enables secure, stateless authentication, while Bcrypt ensures password hashing and security.
- **File Storage: Cloudinary**
  Manages image storage, optimization, and delivery efficiently.
- **AI Integration: Gemini Flash API**
  Powers resume screening and generates intelligent interview questions.

6.CODE AND SCREENSHOTS

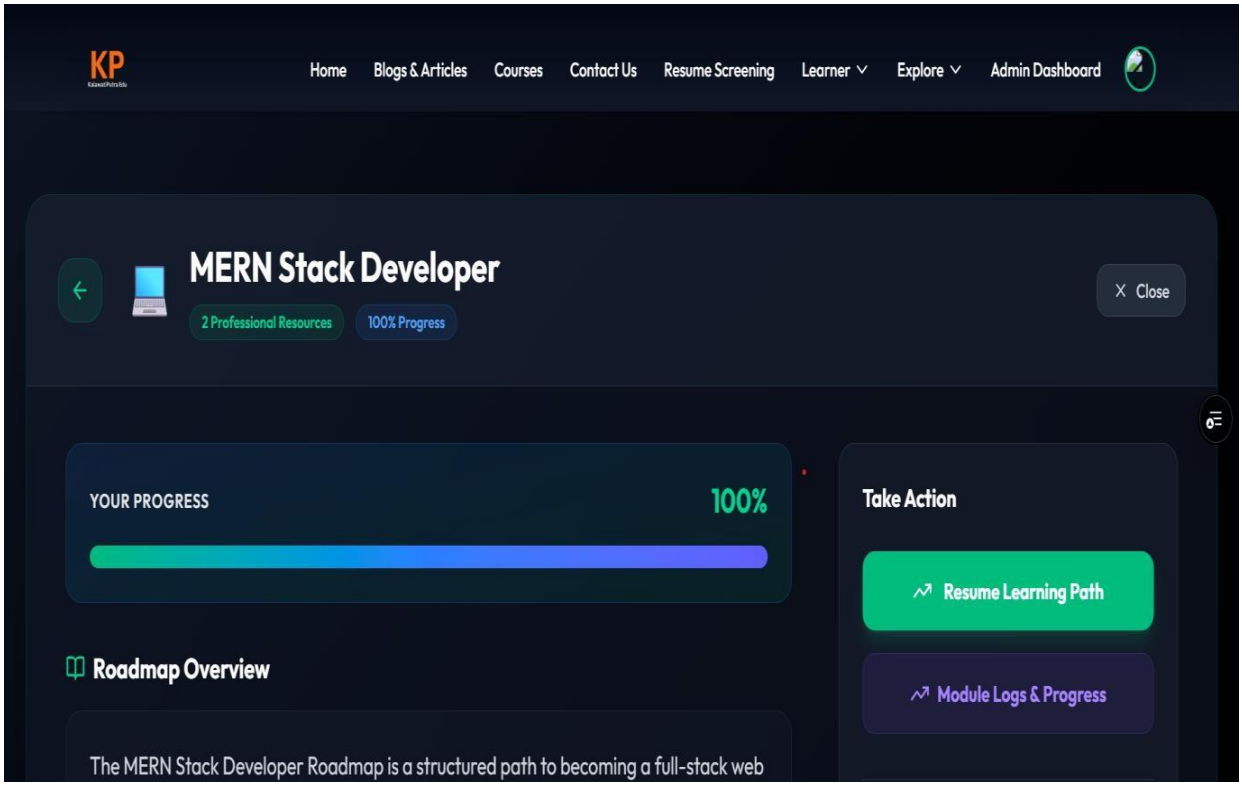
Home Page



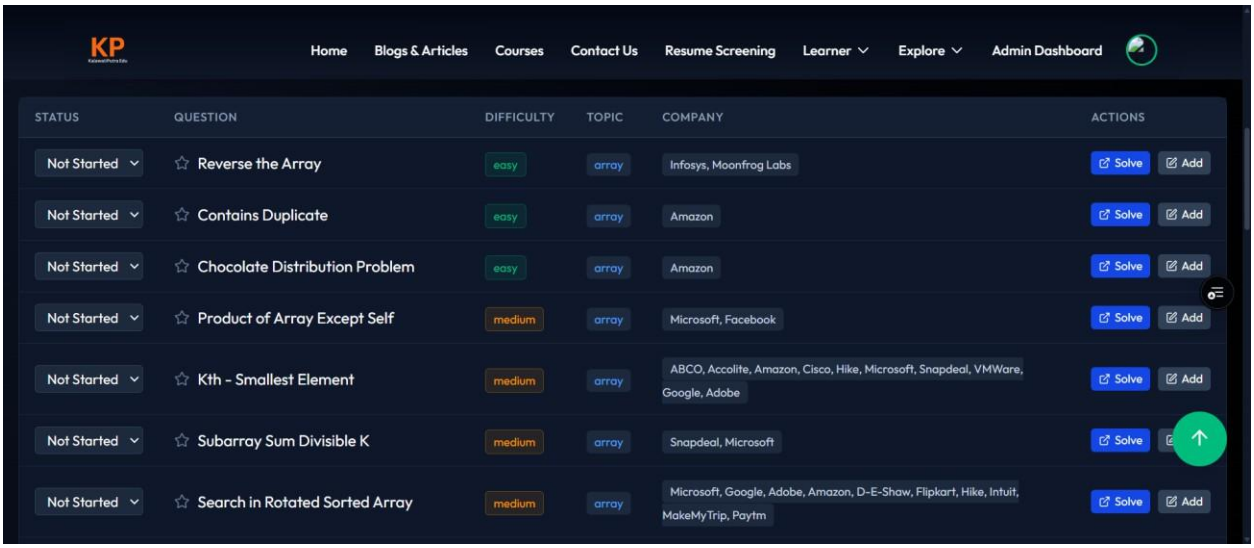
Login – Screen



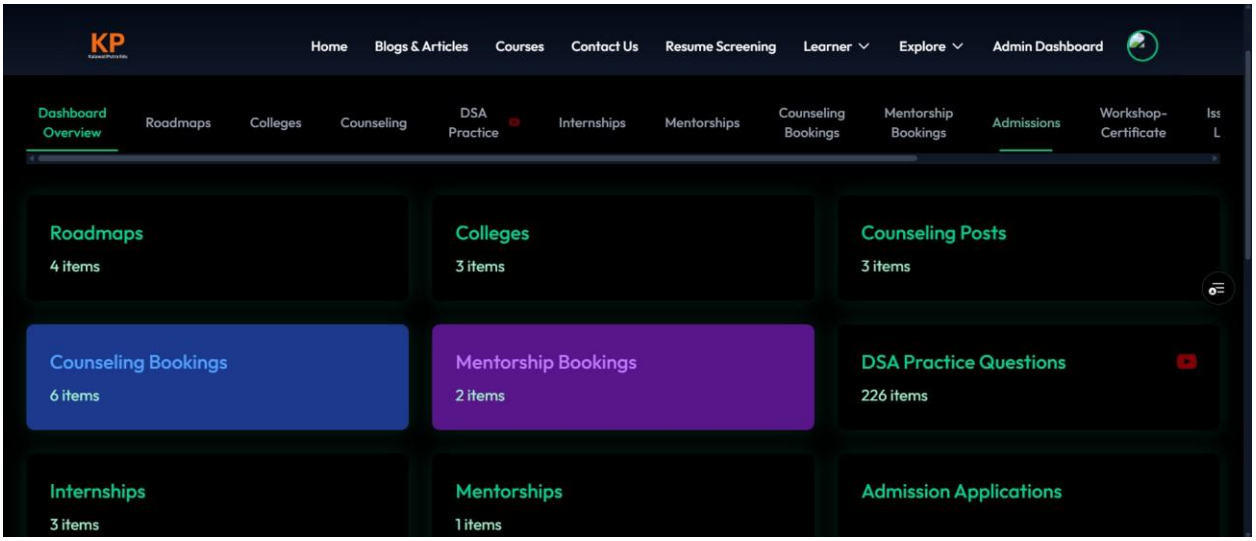
Roadmap


KP
Home
Blogs & Articles
Courses
Contact Us
Resume Screening
Learner
Explore
Admin Dashboard
MERN Stack Developer
2 Professional Resources
100% Progress
X Close
YOUR PROGRESS
100%
Take Action
Resume Learning Path
Module Logs & Progress
Roadmap Overview
The MERN Stack Developer Roadmap is a structured path to becoming a full-stack web

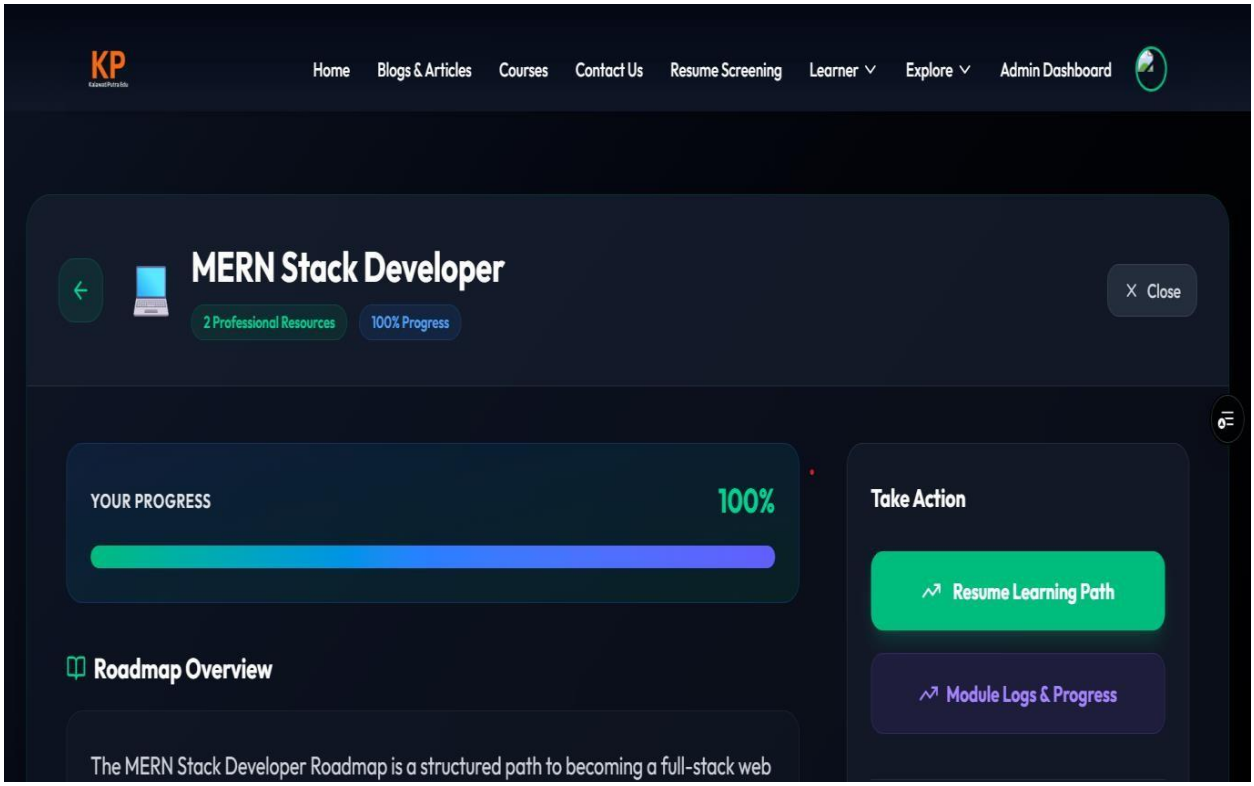
DSA-Practice



Admin



Roadmap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Server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const dotenv = require("dotenv");
const path = require("path");
const passport = require("passport");

const Goodie = require("./models/Goodie");
const Order = require("./models/Order");
const Razorpay = require("razorpay");

dotenv.config();
const app = express();

require("./config/passport");

const cloudinary = require("cloudinary").v2;


// Middleware
// app.use(cors());
app.use(
    cors({
        origin: [
            "https://kalawatiputra.vercel.app",
            "https://kalawatiputra.com",
        ],
    })
);
app.use(express.json());
app.use("/uploads", express.static(path.join(__dirname, "Uploads")));
```

```
32    app.use(express.urlencoded({ extended: true }));
33    app.use(passport.initialize());
34
35    const razorpay = new Razorpay({
36        key_id: process.env.RAZORPAY_KEY_ID,
37        key_secret: process.env.RAZORPAY_KEY_SECRET,
38    });
39
40
41
42
43
44    // Routes
45    const subscribeRoutes = require("./routes/subscribe");
46    const articleRoutes = require("./routes/articles");
47    const courseRoutes = require("./routes/courses");
48    const authRoutes = require("./routes/auth");
49    const adminArticleRoutes = require("./routes/admin/articles");
50    const adminCourseRoutes = require("./routes/admin/courses");
51    const roadmapRoutes = require("./routes/admin/roadmapRoutes");
52    const collegeRoutes = require("./routes/admin/collegeRoutes");
53    const counselingRoutes = require("./routes/admin/counselingRoutes");
54    const dsapracticeRoutes = require("./routes/admin/dsapracticeRoutes");
55    const internshipRoutes = require("./routes/admin/internshipRoutes");
56    const mentorshipRoutes = require("./routes/admin/mentorshipRoutes");
57    const chatbotRoutes = require("./routes/chatbot");
58    const resumeRoutes = require("./routes/resume");
59    const admissionRoutes = require("./routes/admissionRoutes");
60
61    const certificateRoutes = require("./routes/certificate");
```

```
61    const certificateRoutes = require("./routes/certificate");
62    const workshopCertificateRoutes = require("./routes/workshopCertificates");
63    const adminWorkshopRoutes = require("./routes/admin/workshop");
64    const welcomeCertificatesRouter = require("./routes/welcomeCertificates");
65
66    const dsaCertificateRoutes = require("./routes/certificate");
67    const streakRoutes = require("./routes/streak");
68    const issueRoutes = require("./routes/issueRoutes");
69
70    const contactRoutes = require("./routes/contactRoutes");
71
72    const goodiesRoutes = require("./routes/goodies");
73    const cartRoutes = require("./routes/cart");
74    const orderRoutes = require("./routes/orders");
75
76    // Dynamic Sitemap
77    const Article = require("./models/Article");
78    const Course = require("./models/Course");
79
80
81
82    // Use routes
83    app.use("/api/subscribe", subscribeRoutes);
84    app.use("/api/articles", articleRoutes);
85    app.use("/api/courses", courseRoutes);
86    app.use("/api/auth", authRoutes);
87    app.use("/api/users", authRoutes);
88    app.use("/api/admin/articles", adminArticleRoutes);
89    app.use("/api/admin/courses", adminCourseRoutes);
90    app.use("/api/admin/roadmaps", roadmapRoutes);
```

Code Blame 161 lines (131 loc) · 4.97 KB

```
app.use("/api/courses", courseRoutes);
app.use("/api/auth", authRoutes);
app.use("/api/users", authRoutes);
app.use("/api/admin/articles", adminArticleRoutes);
app.use("/api/admin/courses", adminCourseRoutes);
app.use("/api/admin/roadmaps", roadmapRoutes);
app.use("/api/admin/colleges", collegeRoutes);
app.use("/api/admin/counseling", counselingRoutes);
app.use("/api/admin/dsapractice", dsapracticeRoutes);
app.use("/api/admin/internships", internshipRoutes);
app.use("/api/admin/mentorships", mentorshipRoutes);
app.use("/api/chatbot", chatbotRoutes);
app.use("/api/resume", resumeRoutes);
app.use("/api", admissionRoutes);
app.use("/api/certificates", dsaCertificateRoutes);
app.use("/api", contactRoutes);

app.use("/api/goodies", goodiesRoutes);
app.use("/api/cart", cartRoutes);
app.use("/api/orders", orderRoutes);

// app.use('/api/certificates', certificateRoutes);
// app.use('/api/workshop-certificates', workshopCertificateRoutes);
// app.use('/api/admin/workshop', adminWorkshopRoutes);
// app.use('/api/welcome-certificates', welcomeCertificatesRouter);

// Middleware to set certificate type based on route
const setCertificateType = (type) => (req, res, next) => {
    if (!req.query.type) {
        req.query.type = type; // Set type if not provided in query
    }
```

```
120     app.use("/api/certificates", setCertificateType("DSA"), certificateRoutes);
121     app.use(
122         "/api/welcome-certificates",
123         setCertificateType("Welcome"),
124         certificateRoutes
125     );
126     app.use(
127         "/api/workshop-certificates",
128         setCertificateType("Workshop"),
129         certificateRoutes
130     );
131
132     // Admin workshop routes (unchanged)
133     app.use("/api/admin/workshop", adminWorkshopRoutes);
134
135     app.use("/api/auth", require("./routes/auth"));
136     app.use("/api/streak", streakRoutes);
137     app.use("/api/issues", issueRoutes);
138
139     // Serve static files and handle SPA routing
140     if (process.env.NODE_ENV === "production") {
141         app.use(express.static(path.join(__dirname, "../client/dist")));
142         app.get("*", (req, res) => {
143             res.sendFile(path.join(__dirname, "../client/dist", "index.html"));
144         });
145     }
146
147     // MongoDB connection
148     const connectDB = async () => {
149         try {
```

8.CONCLUSION AND FUTURE SCOPE

8.1Conclusion

KalawatiPutra Edu is a **unified ed-tech platform** that successfully solves the problem of fragmented learning resources by integrating:

- DSA practice
- Structured courses
- Progress tracking
- Career services

Key highlights:

- AI-powered resume screening
- Daily goals C streak system
- Admin dashboard for full control

8.2Future Scope

- **Advanced AI Features:** Personalized learning, AI mentor chatbot, blog summaries
- **Online Coding Editor:** Solve and run code directly on the platform
- **Peer Features:** Study groups, peer review system
- **Mobile App:** Android/iOS app for better accessibility

9. REFERENCES

- React – A JavaScript library for building user interfaces.
- Available at: https://reactjs.org/
- Node.js – JavaScript runtime environment.
- Available at: https://nodejs.org/
- Express.js – Web application framework for Node.js.
- Available at: https://expressjs.com/
- MongoDB – NoSQL database for modern applications.
- Available at: https://www.mongodb.com/
- Cloudinary – Media storage and optimization platform.
- Available at: https://cloudinary.com/
- Gemini API – AI tools by Google for developers.

- Available at: https://ai.google.dev/
- JSON Web Tokens (JWT) – Secure authentication method.
- Available at: https://jwt.io/
- LeetCode – Online coding practice platform.
- Available at: https://leetcode.com/
- GeeksforGeeks – Computer science learning platform.
- Available at: https://www.geeksforgeeks.org/

10. APPENDICES

- Project Timeline (Gantt Chart)
- User Manual
- Certificate Design Sample